

Galaxy Quenching and Rejuvenation

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Abstract

To determine what causes galaxies to stop forming stars and what causes “red and dead” galaxies to restart star formation, we first must identify the reason behind specific star formation rate proliferation and reduction. Using a simulation provided by the N-body shop team, I drew a sample of the 40 most massive galaxies in the simulated universe to analyze the root cause of star formation rate fluctuation.

Introduction

First, graphs were needed to see how star formation rate of each specific galaxy changed through its lifetime. A threshold line was plotted which, when star formation rate dips under, signifies low enough star formation rate to be considered ‘non star forming’. This is the natural course a galaxy takes over time, to die out and slow down in star formation, but we frequently see the SFR rise above the line again. To take a look at why this is, I visited the field and plotted the physical galaxies themselves to find out what happened at each point in time and what could have caused the ascensions.

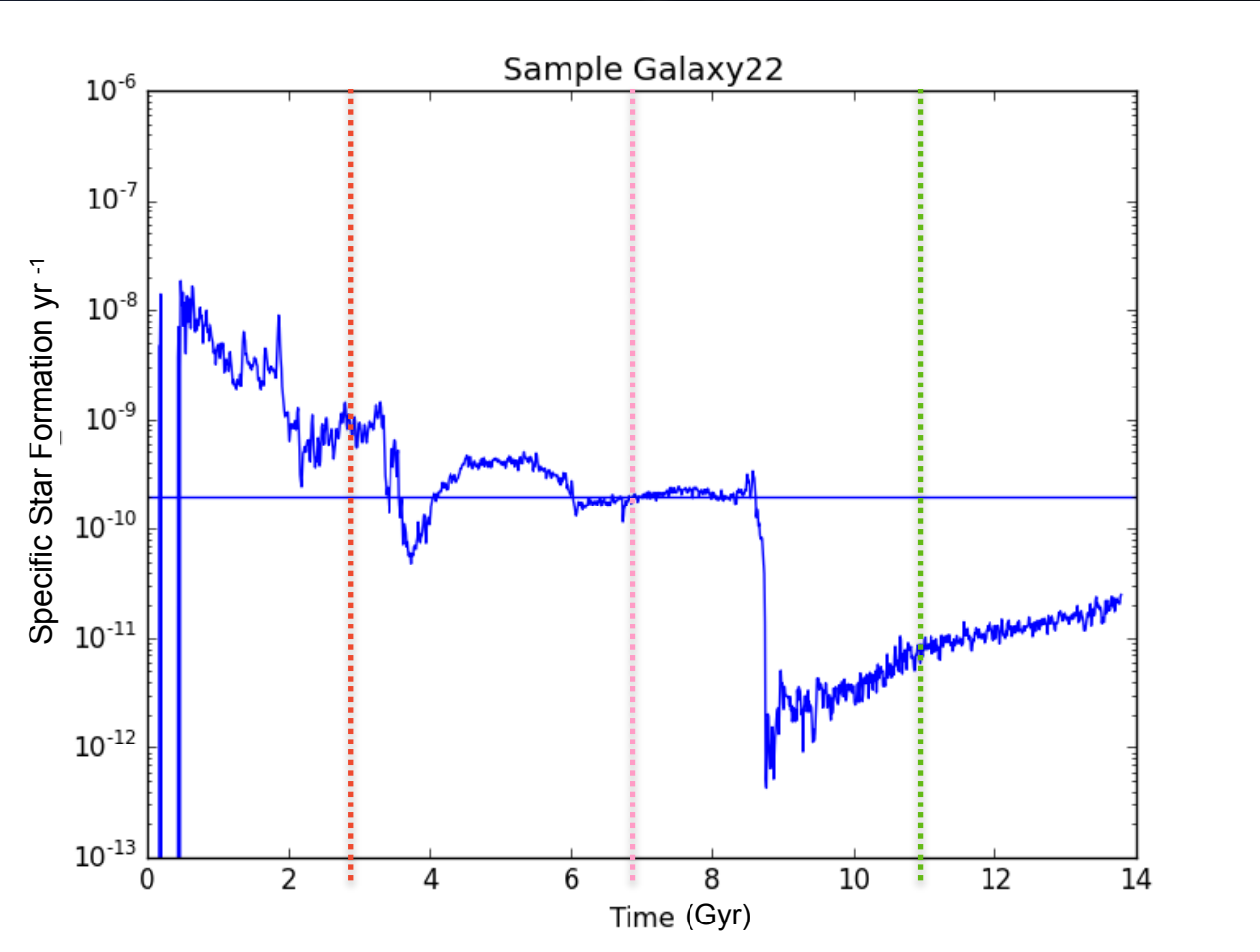
KEY

1945- 3.29 Gyr
4173- 7.04 Gyr
6400- 10.79 Gyr

QUENCHED FROM MERGER
This galaxy starts off as a small, dense, blue galaxy in a cluster of other active, similarly sized blue galaxies. Gas from the nearby halos is likely to interact in the near future.

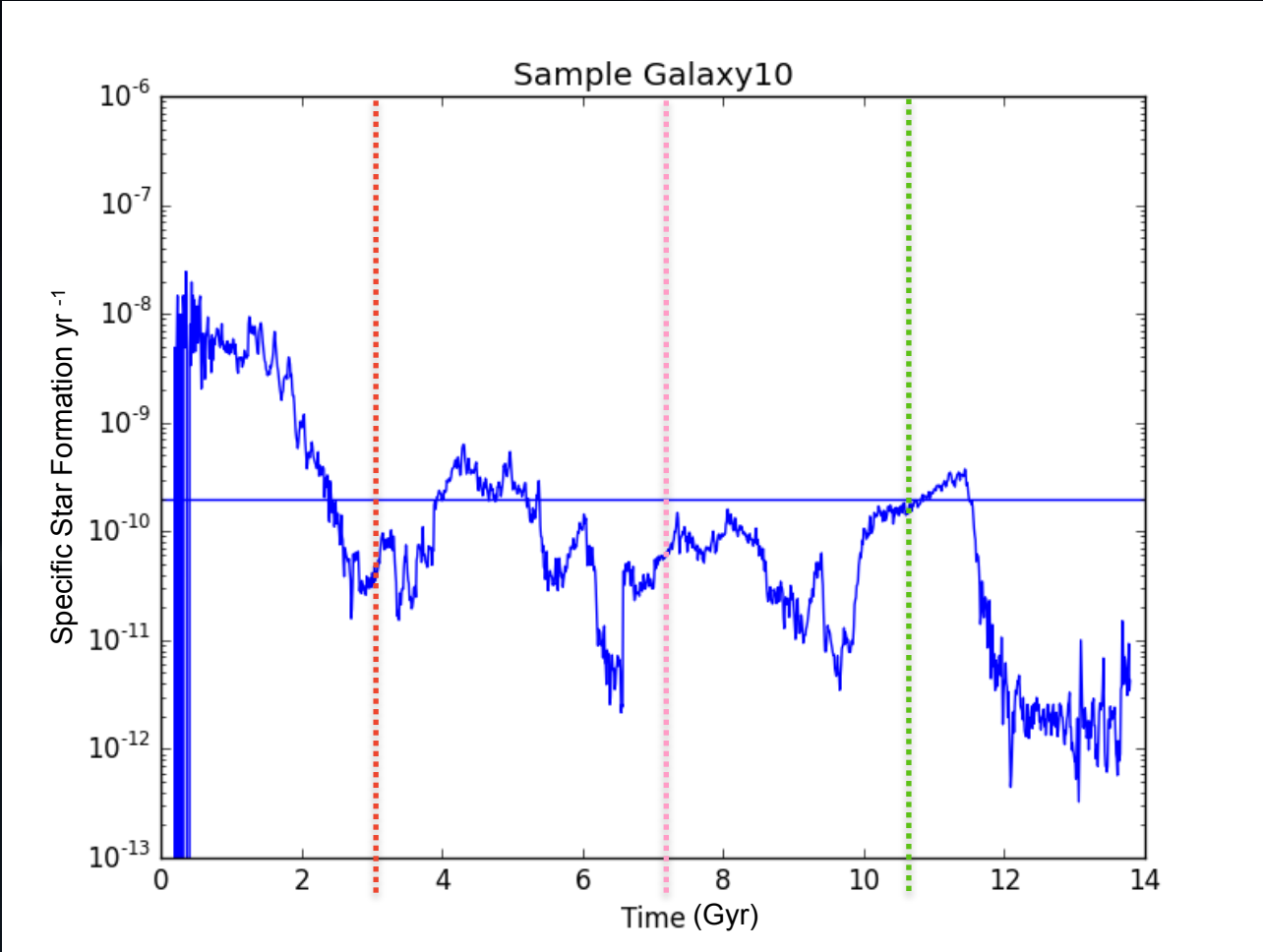
Then the galaxy turns into a giant, dense, blue spiral with only one almost identical neighbor near by.

Lastly, the halo appears to be red and dead with no satellites.



REJUVENATING FROM STRIPPING
This one starts off as a red, low activity, low density galaxy with an active blue neighbor giving off gas. It is below the threshold line at this point.

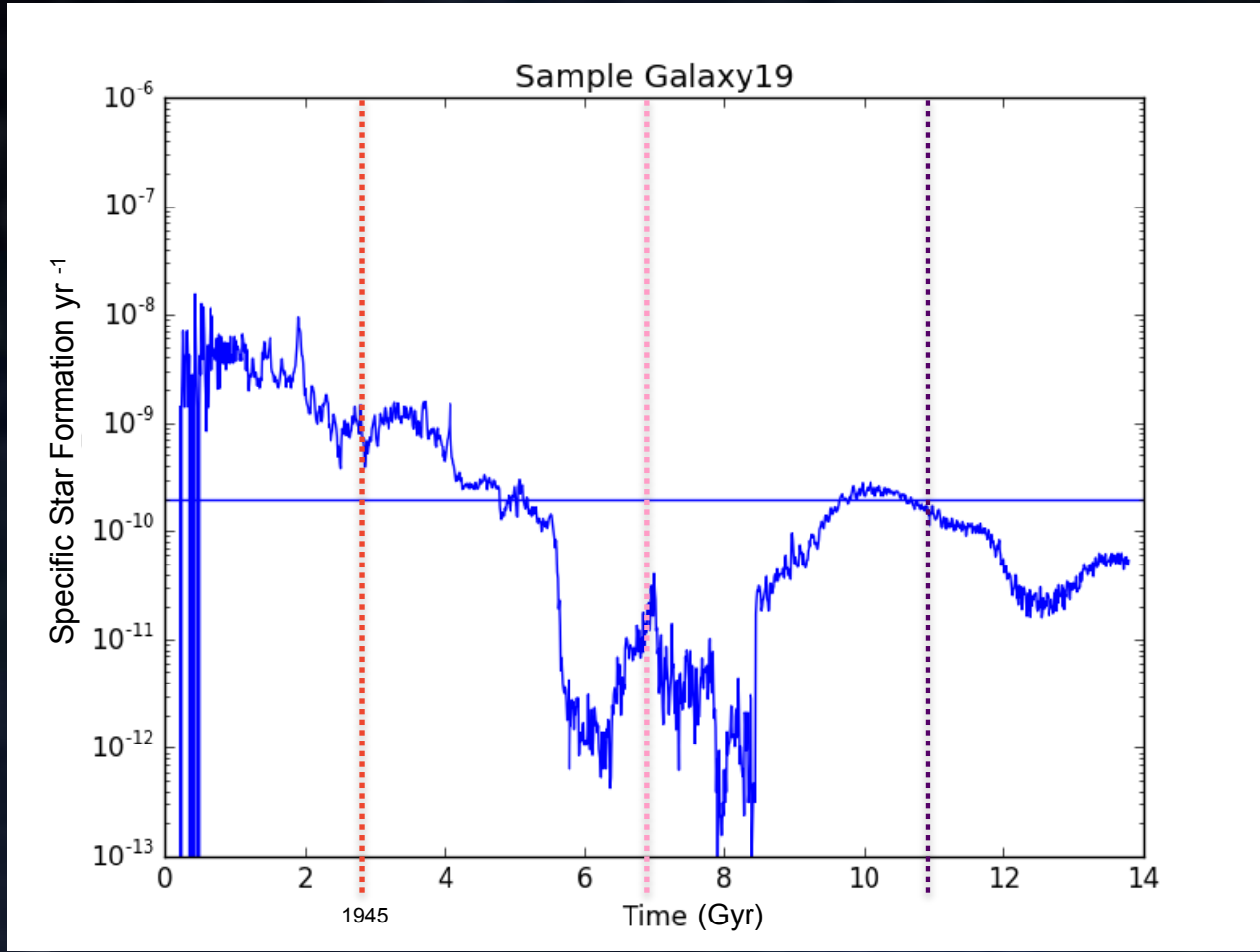
We see the SFR spike up shortly after this and then go back down at 7 Gyr where the galaxy dies and there appears to be remanence of a merger near by. A blue galaxy is close. In the last step the object is a bright, big and blue spiral bar galaxy with very dense gas content. Red and dead satellite near by.



REJUVENATING FROM MERGER
This starts off as a very active, gaseous, blue galaxy. Large dense blue neighbor near by.

Galaxy has turned red and dead, gas depleted. Previously blue neighbor has merged and also died out. Gas source coming from a distant halo.

Lastly the galaxy has rejuvenated likely from the intake in gas from the smaller halo above in the previous step. Extremely dense gas spiral over the red bulge.

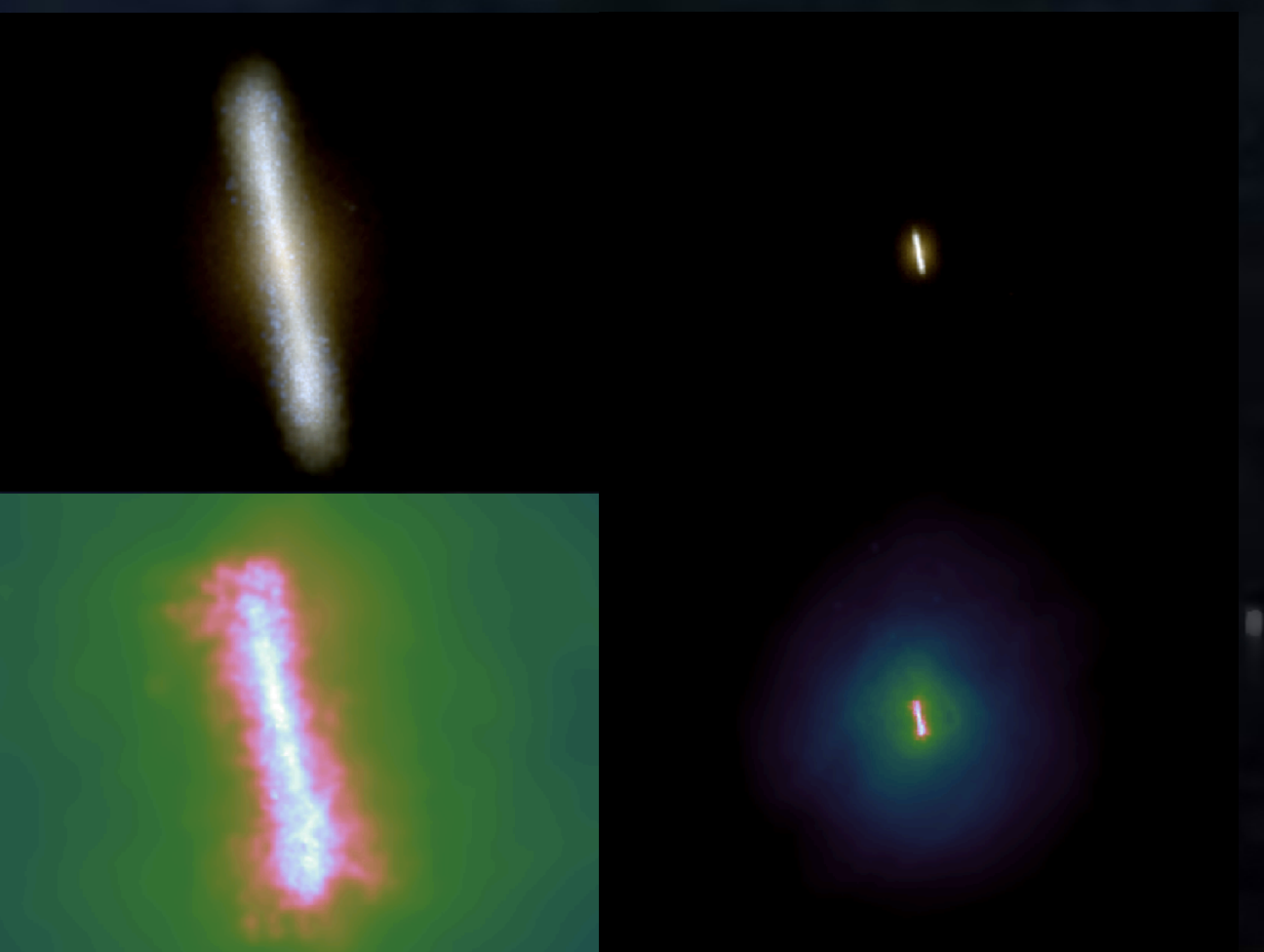
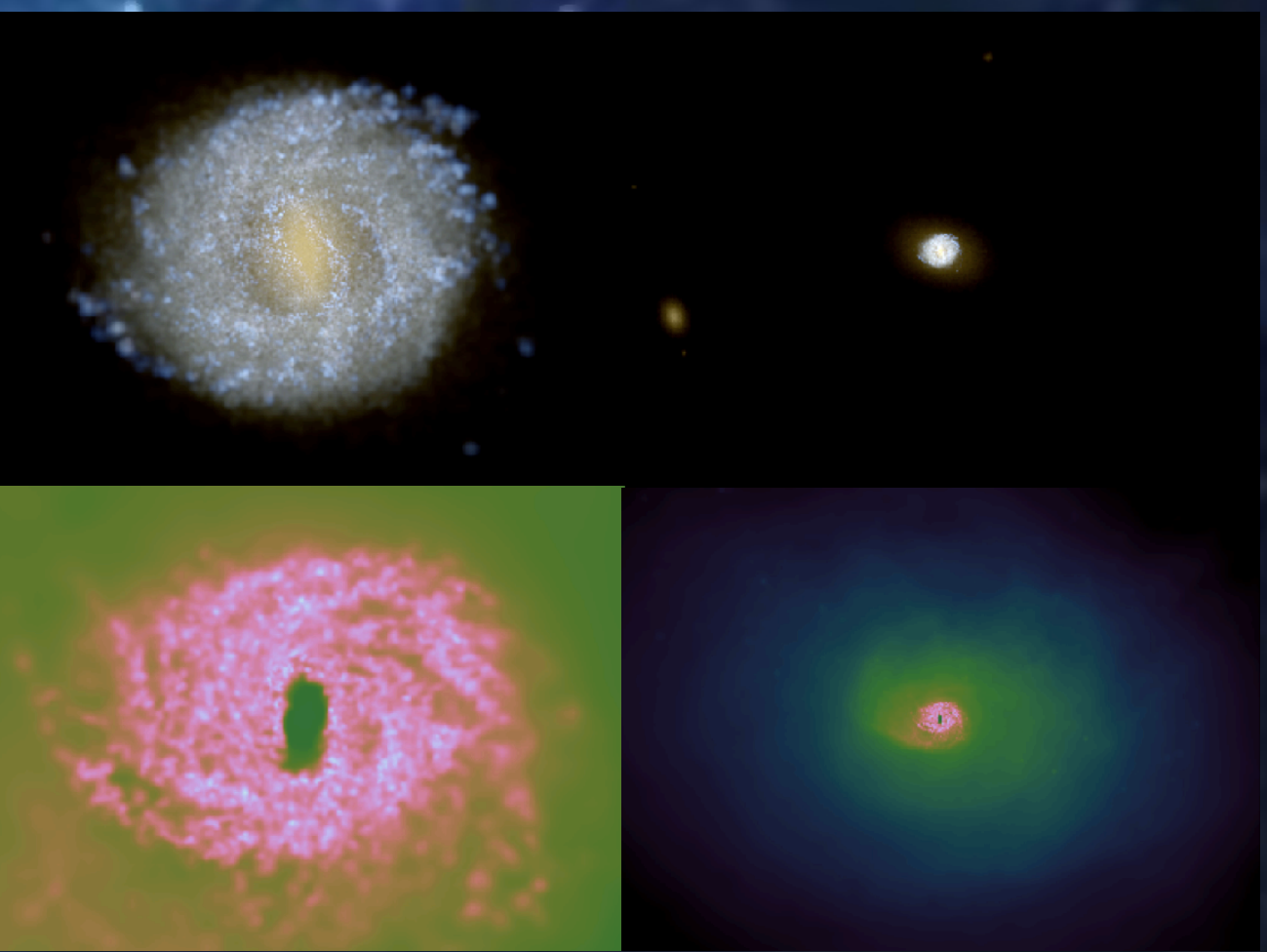
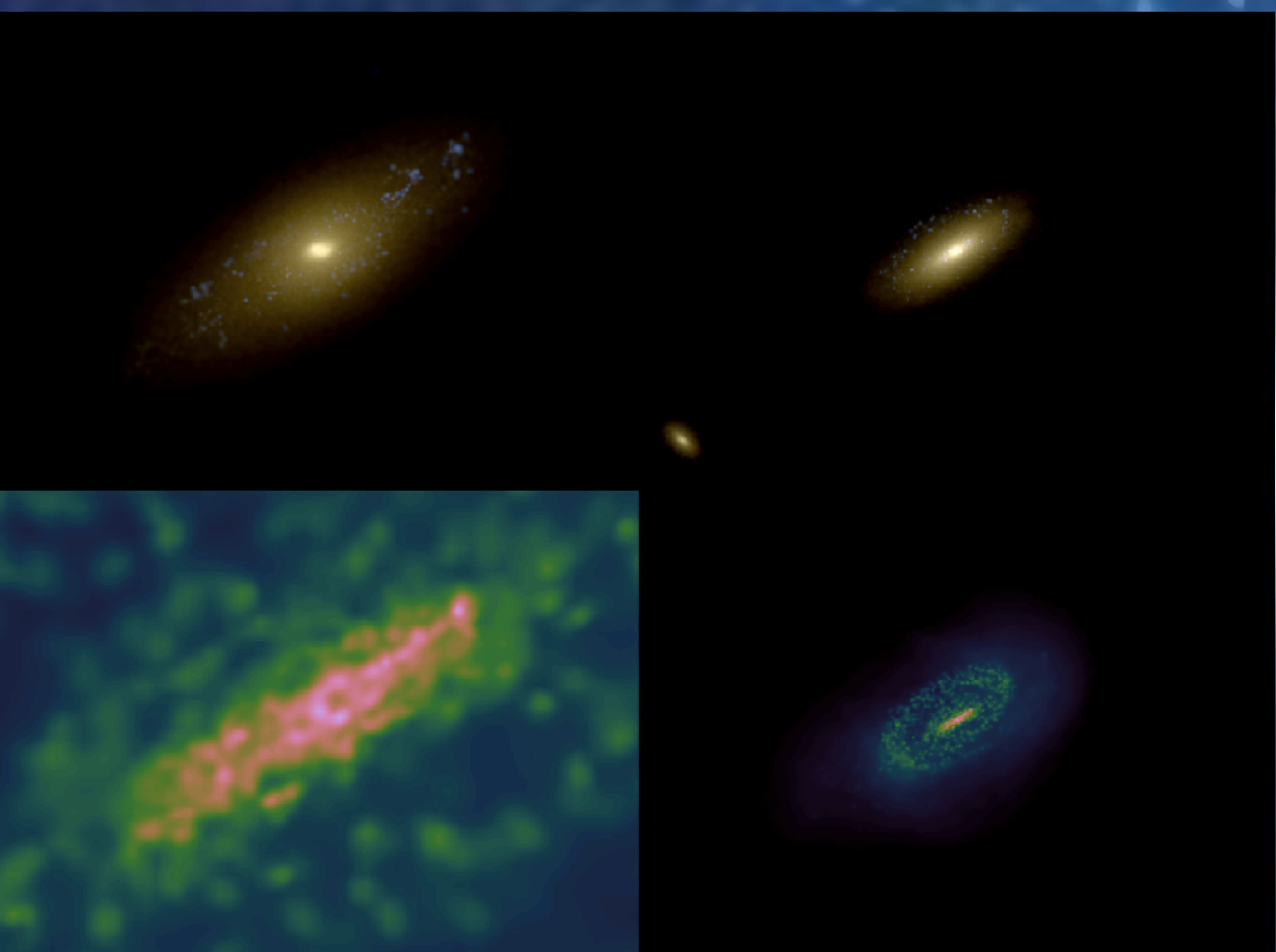
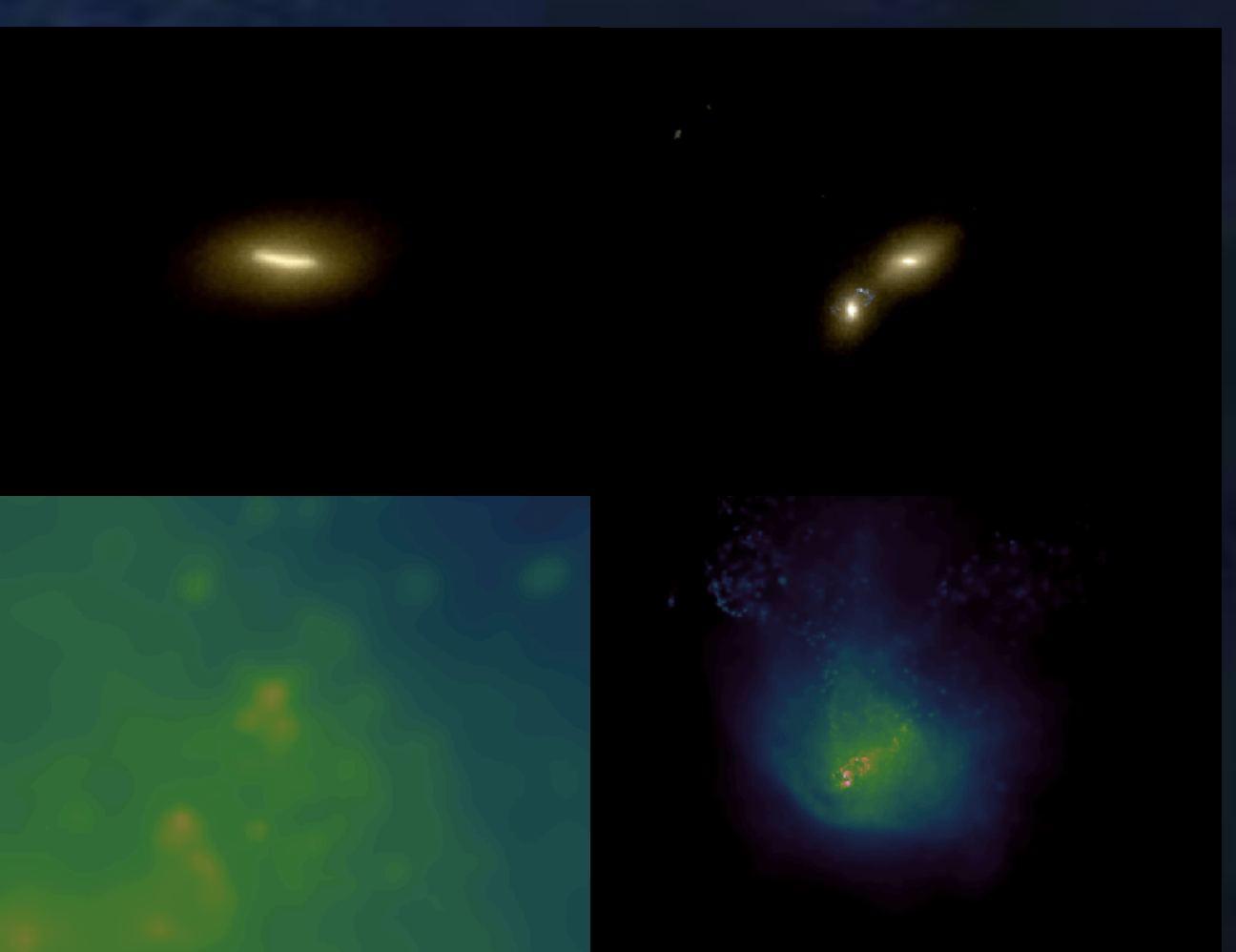
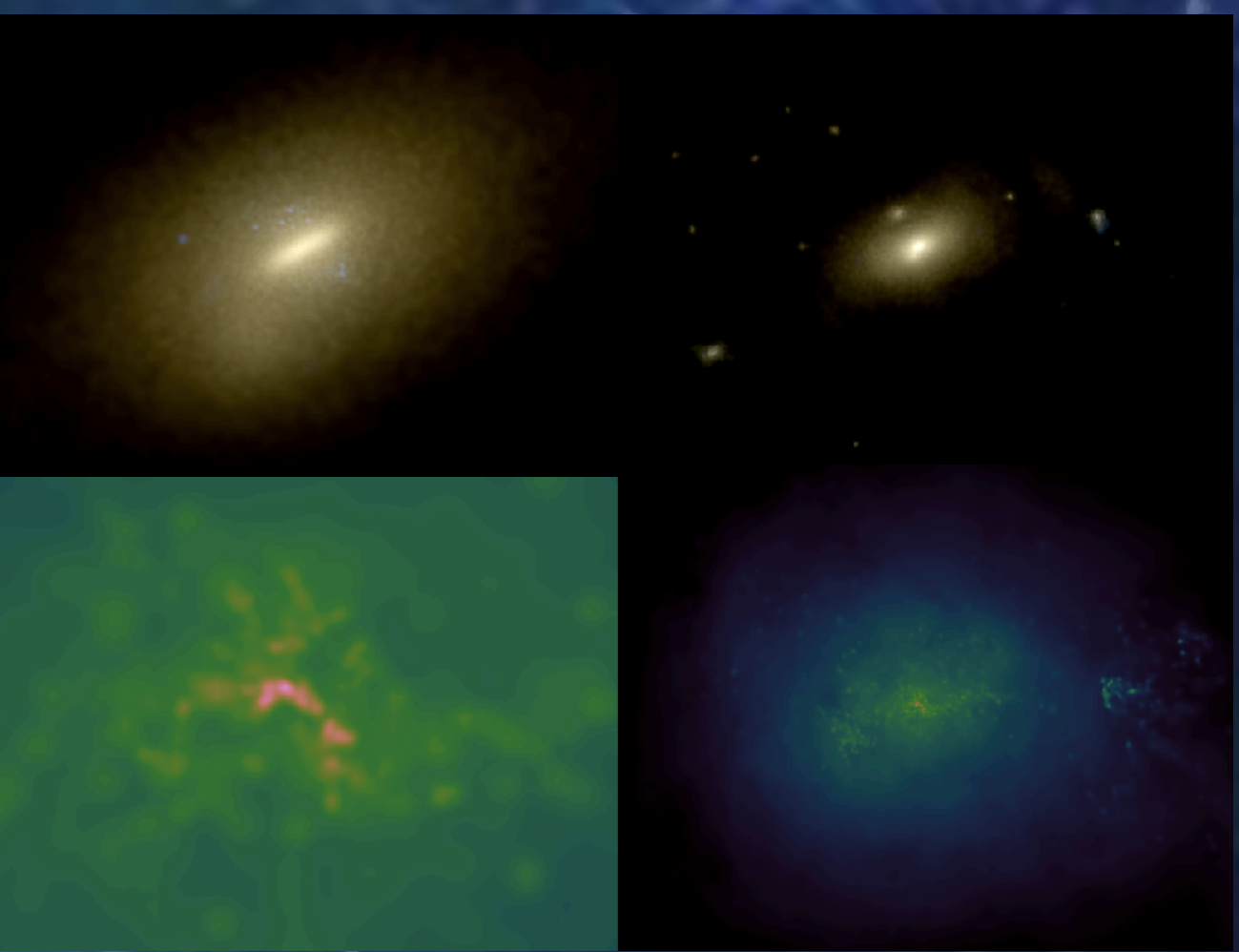
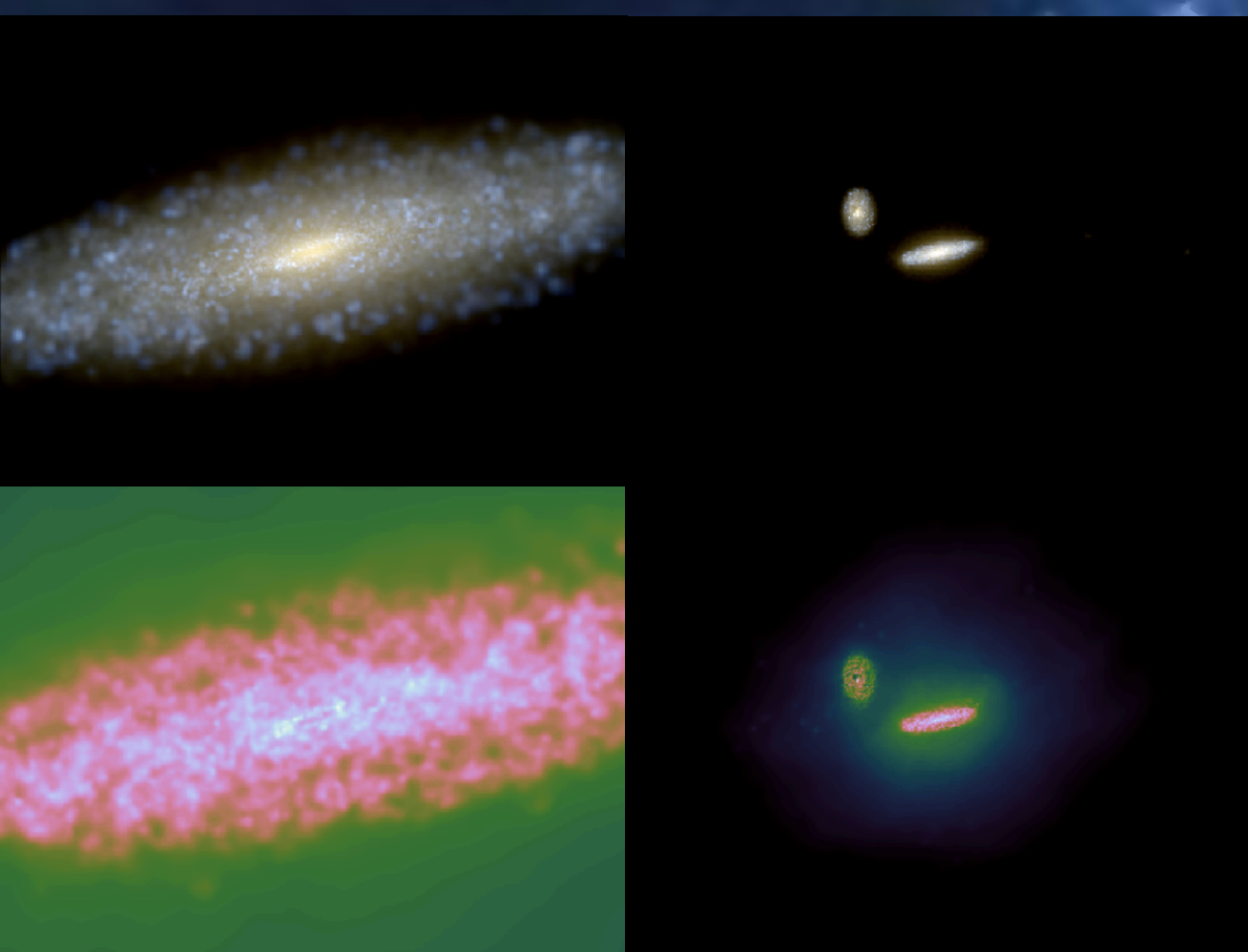
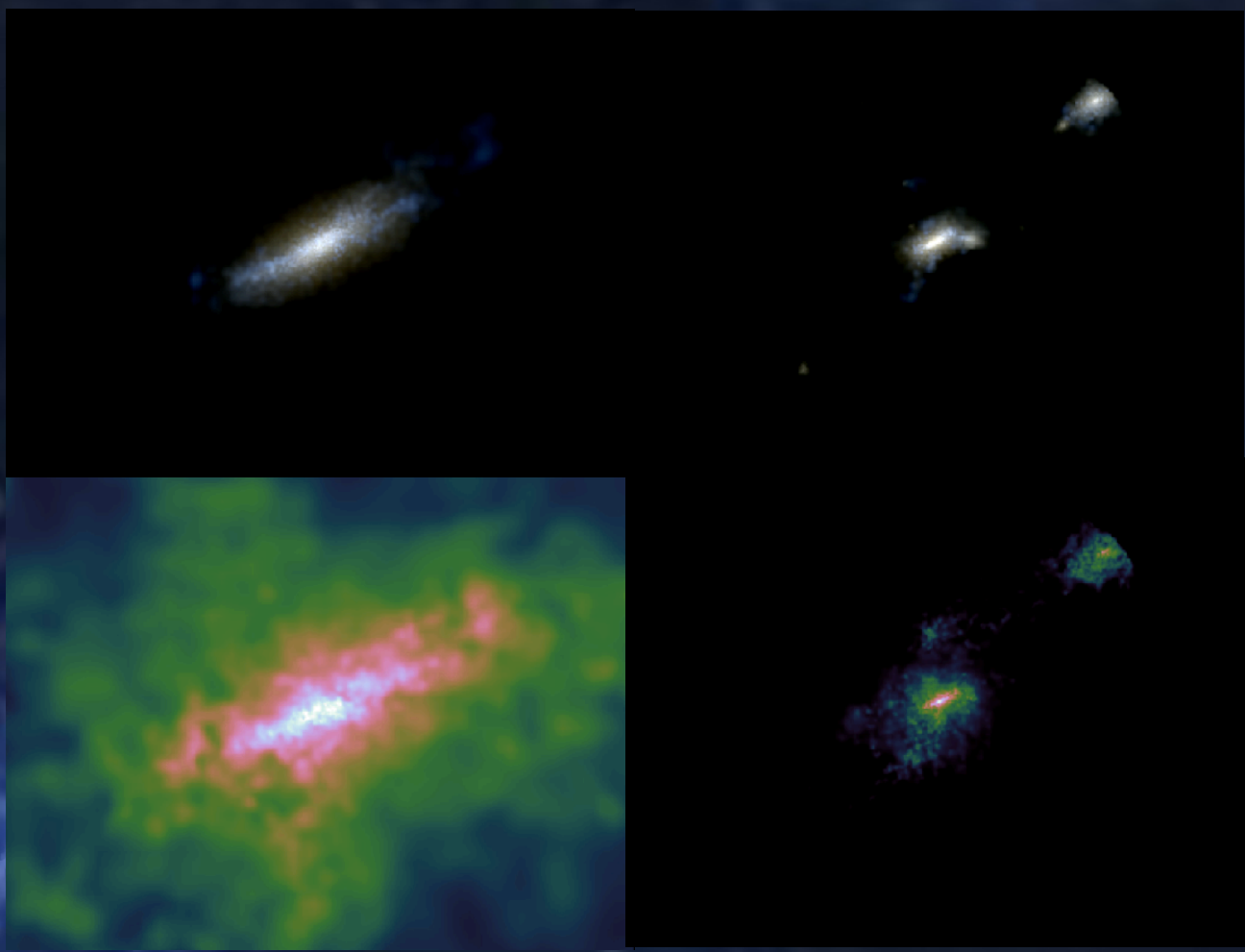
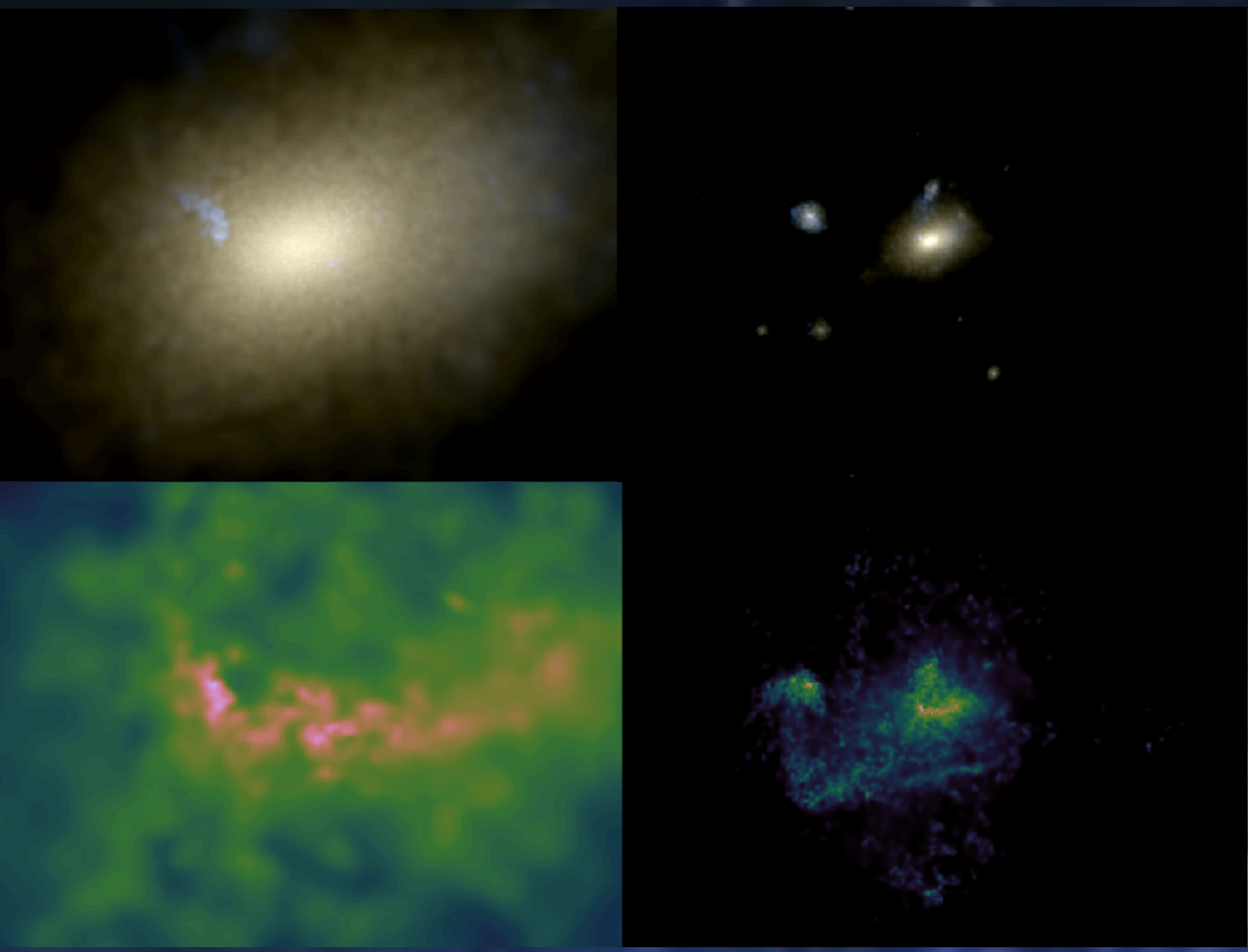
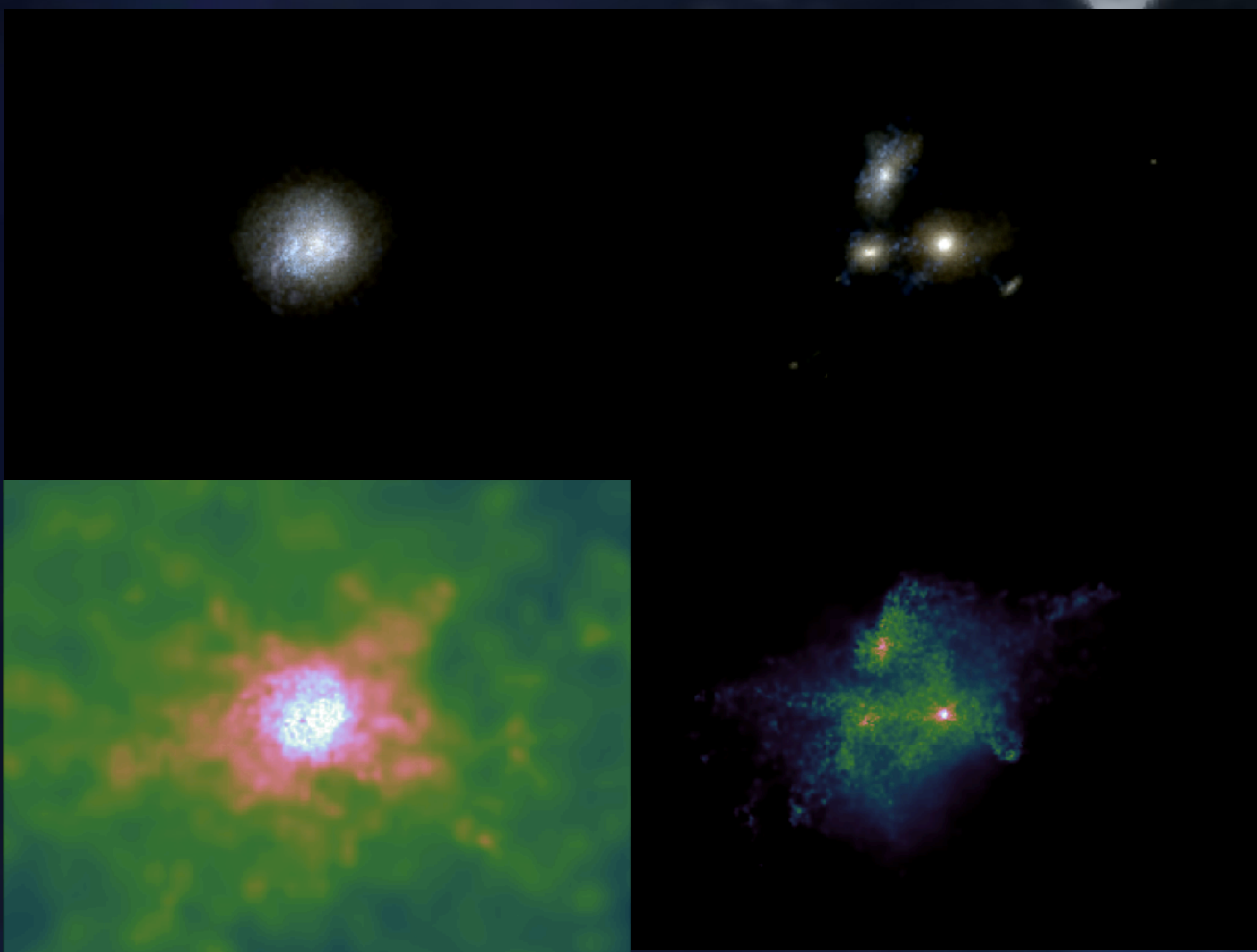


Conclusion

Interactions with other galaxies clearly play a large role in the fluctuation of star formation rate throughout ones lifecycle. With each example of quenching we see a large neighbor close by which is likely to interact in the near future. However, sometimes galaxy interaction results in rejuvenation instead of quenching. It can be observed that the difference is in the mass. Larger galaxy interactions quench galaxies while small ones conversely rejuvenate them. We see examples of small nearby galaxies being stripped of their gas that the larger ones then use to start star forming in its own halo. This process is called rejuvenation by stripping. In other cases, large near by galaxies combine with the observed galaxy to create a merger which results in quiescence.

Future work

Each example of merging gives more information on the nature of each different kind of galaxy interaction. More information is needed to determine the definitive differences in the mergers that cause quenching versus the ones that cause rejuvenation. This knowledge will then be used to understand the role that supermassive black holes play in each case.



Upper left - Galaxy plotted in stars
Upper right - Stars zoomed out
Bottom left - Gas density
Bottom right - Gas density zoomed out

